ModelIt!

Teacher's Guide

Can We Power a City with Wind?

The Science and Challenge of Renewable Energy for Everyone

Quick Reference

Grade Level	Grade 04L2
NGSS Standard	4-ESS3-1, 4-PS3-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-ESS3-1, 4-PS3-1

Obtain and combine information to describe that energy and fuels are derived from natural resources and their uses affect the environment. Use evidence to construct an explanation relating the speed of an object to the energy of that object.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a five-component model showing how wind speed and turbine count interact with electricity generation, city demand, and grid reliability.
Disciplinary Core Idea	**ESS3.A Natural Resources / PS3.A Definitions of Energy** — Energy and fuels humans use are derived from natural resources. Wind is a renewable resource that can be converted to electricity, but its intermittent nature creates challenges for reliable energy supply.
Crosscutting Concept	**Cause and Effect** — Students investigate how changes in wind speed cause predictable changes in electricity generation, and how the relationship between supply and demand determines whether the grid is reliable or fails.

CAST Testing Alignment

MULTIPLE CHOICE:

A city gets all its electricity from a wind farm. On Monday the wind blows strongly, but on Tuesday it is calm. What will most likely happen to the city on Tuesday?

CONSTRUCTED RESPONSE:

Using what you know about wind energy and grid reliability, explain why most cities use a MIX of energy sources instead of relying on wind alone. Use the words 'intermittent,' 'demand,' and 'reliability' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The world is racing to replace fossil fuels with clean, renewable energy sources. Wind power is one of the fastest-growing energy sources on Earth, but it comes with a unique challenge that engineers are still working to solve: the wind does not always blow.

How Wind Turbines Work

A wind turbine converts the kinetic energy of moving air into electrical energy through a surprisingly simple process. Wind pushes the turbine's blades, which are shaped like airplane wings to maximize lift. The spinning blades turn a shaft connected to a generator — a device that converts mechanical (spinning) energy into electrical energy using magnets and copper wire. A single large turbine can generate enough electricity for about 500 homes when the wind is blowing at full speed.

The Intermittency Problem

The biggest challenge with wind energy is intermittency — the wind does not blow at a constant, predictable speed. Wind varies by season, time of day, weather patterns, and geography. A wind farm might produce 100% of a city's needs during a storm and only 5% during a calm day. But the city needs power ALL the time — hospitals cannot lose electricity, refrigerators must keep running, and traffic lights cannot go dark. This mismatch between variable supply and constant demand is the core engineering challenge.

Grid Reliability: Supply Must Match Demand

The electrical grid is like a giant balancing act. Every moment of every day, the total electricity being generated must EXACTLY match the total electricity being used. If supply drops below demand, voltage drops and equipment fails — causing brownouts or blackouts. If supply exceeds demand, equipment can be damaged. Grid operators must constantly adjust generation sources to match demand, which is much harder when a major source (wind) is unpredictable.

Solutions: Batteries, Solar, and Smart Grids

Engineers are developing several solutions to wind's intermittency. Battery storage allows excess electricity from windy days to be saved for calm days — like a rechargeable energy bank. Combining wind with solar energy helps because often when it is not windy, it is sunny (and vice versa). Smart grids use computers to automatically switch between energy sources based on availability. And some regions are building massive pumped-water storage systems where excess electricity pumps water uphill, and when power is needed, the water flows downhill through turbines to generate electricity.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Wind Speed and Number of Turbines as external components (environmental conditions and human decisions) and Electricity Generated, City Energy Demand, and Grid Reliability as internal components.

E **Establish Relationships**

Students determine that Wind Speed and Number of Turbines both positively affect Electricity Generated, City Energy Demand negatively affects Grid Reliability (more demand strains the system), and Electricity Generated positively affects Grid Reliability (more supply stabilizes it).

V **Visualize & Model**

Students build a five-component model showing how two inputs (wind and turbines) create electricity that must compete with city demand to determine whether the grid stays reliable.

E **Evaluate Outcomes**

Students run four scenarios to observe how wind variability creates reliability problems even with many turbines, discovering that adding turbines helps but cannot solve calm-day shortages.

R **Revise & Extend**

Students add battery storage or solar panels to explore how combining energy sources and storing excess electricity can stabilize the grid and reduce blackout risk.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"The wind is free. It is everywhere. It never runs out. So why can we not just use it to power every city on Earth?"*

Watch For

Show images of wind farms. Ask: If wind is free, why does your city still burn coal and gas? What is the problem?

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to build a model that shows the REAL challenge of wind energy — and it is not what you think."*

Watch For

Have students read objectives aloud. Pre-teach 'intermittent' and 'grid reliability' with simple examples.

Slide 3: Big Question

3 min

Say: *"Here is the question every energy engineer is trying to answer: can wind replace all the coal, gas, and oil that power our cities?"*

Watch For

Have students vote: do you think wind can power a city by itself? Write predictions and reasoning.

Slide 4: LEVER Framework

2 min

Say: *"We are modeling an ENTIRE energy system today — from the wind blowing to the lights turning on in your house."*

Watch For

Preview all five components. Explain that this model connects nature (wind) to engineering (turbines) to society (city demand).

Slide 5: Activity 1: Components

8 min

Say: *"Which of these can humans control? We cannot control the wind, but we CAN choose how many turbines to build."*

Watch For

Guide sorting. Wind Speed is external (nature controls it). Number of Turbines is external (humans choose). The three internals are results.

Slide 6: Activity 2: Connections

10 min

Say: *"Look carefully — Energy Demand has a NEGATIVE effect on Grid Reliability. Why? Because more demand without more supply creates problems."*

Watch For

Help students map all four relationships. Discuss why supply must match demand and what happens when it does not.

Slide 7: Activity 3: Simulation

12 min

Say: *"Watch Grid Reliability when we change the wind. On a windy day, we are golden. But what happens when the wind drops?"*

Watch For

Run all four scenarios. The calm-day scenario is the key moment — students see grid reliability crash even with many turbines.

Slide 8: Discoveries

5 min

Say: *"Wind energy is incredible — it is clean, free, and renewable. But it has one big weakness. Can anyone name it?"*

Watch For

Class discussion: What did the model show us about wind's biggest limitation? Introduce the word 'intermittent' formally.

Slide 9: STEM Challenge

12 min

Say: *"Your challenge: design a community energy plan that uses MOSTLY wind but has a smart backup plan for calm days."*

Watch For

Groups design energy plans on poster paper: how many turbines, where, and what backup. Present and defend with model evidence.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Wind Speed, Number of Turbines
- **INTERNAL:** Electricity Generated, City Energy Demand, Grid Reliability

Relationships

- **Wind Speed to Electricity Generated:** POSITIVE (+) — Faster wind spins turbine blades faster, generating more electricity. Wind energy increases with the cube of wind speed — doubling wind speed actually produces eight times more energy. This is why wind farms are built in the windiest locations possible.
- **Number of Turbines to Electricity Generated:** POSITIVE (+) — More turbines capture more wind energy. A wind farm with 100 turbines generates roughly five times more electricity than one with 20 turbines. Building more turbines is the most direct way to increase total generation capacity.
- **City Energy Demand to Grid Reliability:** NEGATIVE (-) — Higher energy demand puts more strain on the grid. If demand increases but supply stays the same, the gap between what is needed and what is available grows, making blackouts more likely. Cities with higher demand need more generation to maintain reliability.
- **Electricity Generated to Grid Reliability:** POSITIVE (+) — More electricity generation means more supply available to meet the city's demand. When generation exceeds demand, the grid is stable and reliable. The closer generation gets to matching or exceeding demand, the more reliable the grid becomes.

Key Discoveries

- Wind turbines convert the kinetic energy of moving air into electrical energy — faster wind produces more electricity
- Grid reliability depends on SUPPLY matching DEMAND — if the wind drops and generation falls below what the city needs, blackouts happen
- Wind energy is intermittent (not constant) — this is the biggest challenge, because cities need power 24/7 but wind does not blow 24/7
- Adding more turbines helps generate more electricity on windy days, but cannot solve the fundamental problem of calm days

Multiple Choice Post-Assessment Answer Key

1. Wind turbines convert one form of energy into another. What energy transformation happens inside a wind turbine?

Correct: B) The kinetic energy of moving air is converted into electrical energy

That is right! Wind turbines capture the kinetic energy (motion energy) of the wind. As the blades spin, generators inside convert that motion into electrical energy that can power homes and schools.

2. In the model, Wind Speed and Number of Turbines are external components. Why are these considered external?

Correct: B) Because they are conditions or design choices that determine how the rest of the system behaves

Correct! External components are the inputs we set. Wind Speed is the environmental condition, and Number of Turbines is the design choice. The internal components (Electricity Generated, City Energy Demand, Grid Reliability) respond to these inputs.

3. The model shows that City Energy Demand has a NEGATIVE relationship with Grid Reliability. What does this mean for a city?

Correct: B) When City Energy Demand increases but electricity supply stays the same, Grid Reliability decreases because demand outpaces supply

Exactly! A negative relationship means when one goes up, the other goes down. If the city needs more power but the turbines are not producing enough, the gap between supply and demand grows, making blackouts more likely.

4. A student runs the model with Wind Speed at 15% and Number of Turbines at 80%. Grid Reliability drops very low. What does this tell us about wind energy?

Correct: C) Wind energy is intermittent, so even many turbines produce very little electricity on calm days, causing the supply to fall below the city's demand

Right! This is the core challenge of wind energy. Wind is intermittent, meaning it does not blow constantly. Even 100 turbines produce almost nothing on a calm day, but the city still needs power for hospitals, schools, and homes.

5. Based on the model, why can wind energy alone NOT reliably power a whole city?

Correct: C) Wind speed changes constantly, so electricity generation goes up and down while the city's demand stays high, causing blackouts on calm days

That is right! The model shows that Grid Reliability depends on supply matching demand. Wind speed is unpredictable and intermittent, so generation fluctuates. On calm days, supply drops far below demand. That is why wind works best as part of an energy mix with solar, batteries, or other sources.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Wind turbines work all the time as long as they are built"	Wind turbines only generate electricity when the wind is blowing above a minimum speed (usually about 7-9 mph). On calm days, they produce almost nothing. Even in the windiest locations, turbines typically operate at full capacity only about 25-45% of the time. The rest of the time, they are either idle or running at reduced output.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We just need to build more turbines to solve the energy problem"	Adding more turbines increases capacity on WINDY days but does nothing to help on CALM days. If the wind is not blowing, it does not matter if you have 10 turbines or 10,000 — they all produce zero electricity. The solution is not just more turbines but STORAGE (batteries) and DIVERSITY (multiple energy sources). Our model showed that even at 100% turbines, a calm day crashes grid reliability.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Wind energy is too unreliable to use"	While wind alone cannot power a city reliably, it is an essential part of the energy mix. Wind already provides about 10% of US electricity and over 50% in some countries like Denmark. The key is combining wind with solar, batteries, and other sources. Wind's variability is a challenge, not a dealbreaker — smart engineering solutions are making wind energy more reliable every year.